

ANDRÉ CORREIA

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SUMMARY

Machine Learning Engineer & Researcher with a PhD and 5+ years of combined R&D and industry experience in Computer Vision, Imitation & Reinforcement Learning, and LLMs. Track record across the entire ML lifecycle, from research and algorithm design to deployment in retail, industry, airport, and surveillance domains.

SKILLS & LANGUAGES

Machine Learning	Imitation, Reinforcement and Supervised Learning, Computer Vision, LLMs
Tools & Frameworks	PyTorch, TensorFlow, NumPy, Pandas, Scikit-Learn, Flask, FastAPI, Docker, Git
Programming	Python, Java, C++, C, C#
Languages	Native Portuguese, C2 English, B1 Spanish, A2 German

WORK EXPERIENCE

DeepNeuronic	December 2024 - Present
<i>Machine Learning Engineer and Researcher</i>	

- Delivered dozens of new custom computer vision solutions specific to airport management within a critical 2-week deadline, successfully winning the open project and establishing a major long-term partnership.
- Created a custom theft action dataset and benchmarked 200+ Large Video Language Models (LVLMs) with prompt engineering for theft detection in surveillance applications. The resulting model has 82% accuracy.
- Researched state-of-the-art multi-camera tracking and Person Re-ID methods. Combined baseline algorithms to enhance tracking robustness across multiple camera views. It is now the primary tracker used by the company.
- Built a large-scale age and gender dataset combining public data, manual annotation, and synthetic data via generative models; trained a YOLO classification model achieving 87% accuracy and deployed it via Docker.
- Created and annotated a personal protective equipment (PPE) dataset and trained a YOLO model for real-world PPE detection, achieving a 98% mAP score and deployed it via Docker to manage PPE compliance in industry.
- Identified performance bottlenecks and refactored the core inference pipeline, significantly reducing latency and vulnerabilities while improving code quality.

PROJECTS AND PUBLICATIONS

Hierarchical Decision Transformer and Mamba	2023 & 2026
Advanced the SOTA sequence modeling in offline RL using Transformer and Mamba models.	Link — Link
Music to Dance as Language Translation	2026
Transformer and Mamba based framework to translate music to dance, applied to a robotic arm.	Link
A Survey of Demonstration Learning	2024
Survey covering problem formulation, learning methods, benchmarks, and open research challenges.	Link
DEFENDER	2023
Algorithm to improve RL safety during training using limited demonstrations.	Link
Multi-View Contrastive Learning from Demonstrations	2022
Algorithm to learn task-relevant features from demonstration to train context-independent policies.	Link

EDUCATION

Universidade da Beira Interior, Covilhã	
PhD in Computer Science (Engineering) - High Honors Dissertation: Improving the Robustness of Demonstration Learning	August 2020 – October 2024
Bsc and MSc in Computer Science (Engineering) - CGPA: 18/20	August 2015 – June 2020